

SSD_Power Adjustment Reference

1. Turn off the analog circuit

Turn off the following options in the kernel config (Note: mark * to turn it off. If you need to click the screen to have UI display, DISP_LPLL/UPLL_0/UPLL_1 cannot be turned off; if you need to click on the mipi screen, MIPI_DPHY_TX cannot be turned off.)

```
[*] Do not support debugfs for clock platform
[*] Skip squashfs bad block
[*] support deferred initcalls
[ ] defer slab_sysfs_init
[ ] defer param_sysfs_init
[ ] defer perf_event_sysfs_init
[ ] defer irq_sysfs_init, init_clocksource_sysfs, clockevents_init_sysfs
[ ] defer of_core_init
[ ] Do not create crypto test thread
[*] SoC infinity2m (ARCH_MULTI_V7)
[ ] infinity2m FPGA environment
[*] Power down audio at boot
[*] Power down EMAC at boot
[*] Power down HDMI_ATOP at boot
[*] Power down IDAC_ATOP at boot
[*] Power down IDAC_LPLL at boot
[*] Power down DISP_LPLL at boot
[*] Power down MIPI_DPHY_TX_TOP at boot
[*] Power down SATA_ATOP at boot
[*] Power down UPLL_0 at boot
[*] Power down UPLL_1 at boot
[*] Power down USB20_P1 at boot
[*] Power down USB20_P2 at boot
[*] Power down USB20_P3 at boot
[ ] Record timestamp in sram
```

2. Turn off the corresponding digital circuit

- The function modules corresponding to Audio & HDMI & IDAC(VGA) & DISP are configured in the SDK, and the corresponding SDK modules can be turned off:

project\release\customer_tailor\nvr_i2m_display_glibc_tailor.mk

- audio

```
interface_ai:=disable
interface_ao:=disable
mhal_aio:=disable
```

- hdmi

```
interface_hdmi:=disable
mhal_hdmity:=disable
```

- disp

```
interface_disp:=disable
mhal_disp:=disable
```

- mipi tx

```
interface_panel:=disable
mhal_panel:=disable
```

- EMAC网络开关的config在kernel的config里面：

```
--- SStar SoC platform drivers
[ ] SStar Cam drivers ----
[ ] SStar DLA driver ----
[ ] PIU Timer driver
[*] msys api
[ ] log Redirection
[ ] sysfs:perf test
[ ] ioctl:bench memory function
[ ] sysfs:MIU protest
[ ] add more dmem sysfs node
[*] Serial / UART driver
< * MS_USCLK extended sysfs interfaces for CLKs
[*] Serial Flash driver
[*] Customized ZY Erase ENV
< * SSTAR_PWM
< * SPI NAND
< * MS SPI driver
< * MMC driver
< * SStar SD/MMC Card Interface Support
< * EMAC
< * Crypto driver
[ ] Enable AES DMA interrupt
< * Support cryptodev
< * Mstar CPU frequency scaling driver
-* MSTAR MIU
!(+)
```

```
Linux/arm 4.9.84 Kernel Configuration
submenus ---> (or empty submenus ----). Highlighted letters are hotkeys. Pressing <Y> i
h. Legend: [*] built-in [ ] excluded <M> module < > module capable

-* Patch physical to virtual translations at runtime
  General setup --->
[*] Enable loadable module support --->
[*] Enable the block layer --->
  System Type --->
  Bus support --->
  Kernel Features --->
  Boot options --->
  CPU Power Management --->
  Floating point emulation --->
  Userspace binary formats --->
  Power management options --->
[ ] Networking support ----
  Device Drivers --->
  Firmware Drivers --->
  File systems --->
  Kernel hacking --->
  Security options --->
[*] Cryptographic API --->
  Library routines --->
[ ] Virtualization ----
```

```
SStar SoC platform drivers
submenus ---> (or empty submenus ----). Highlighted letters are hotkeys. Pressing <Y> i
Legend: [*] built-in [ ] excluded <M> module < > module capable

r(-)
< > EMAC
<M> Crypto driver
[ ] Enable AES DMA interrupt
<M> Support cryptodev
<*> Mstar CPU frequency scaling driver
-* MSTAR MIU
  MIU_PROTECT (ssc007a-s01a) --->
[*] Mstar BDMA driver
[ ] BDMA Line-Offset Enable
[*] Mstar Move DMA driver
< > SStar SATA HOST
< > SStar IR driver
<*> Sstar I2C driver
[ ] Support I2c ISR
<*> GPIO driver
< > SW I2C via GPIO support
[*] MS_RTC
< > Mstar RTC driver
<*> Mstar RTCWC driver
[ ] Mstar RTCPWC driver Error Handle on Hw Exception
<*> watchdog driver
< > sar driver
[*] MMA_HEAP
<> SSTAR 10/100 PHYs
<*> SStar Voltage Control
```

- Sata开关的config在kernel的config里面：

```

Generic Driver Options --->
Bus devices --->
< > Connector - unified userspace <-> kernelspace linker ----
-* Memory Technology Device (MTD) support --->
-* Device Tree and Open Firmware support --->
< > Parallel port support ----
[ ] Block devices ----
< > NVMe Target support
Misc devices --->
SCSI device support --->
< > Serial ATA and Parallel ATA drivers (libata) ----
[ ] Multiple devices driver support (RAID and LVM) ----
< > Generic Target Core Mod (TCM) and ConfigFS Infrastructure ----
[*] Network device support --->
[ ] Open-Channel SSD target support ----
Input device support --->
Character devices --->
I2C support --->
[*] SPI support --->
< > SPMI support ----
< > HSI support ----
PPS support --->
PTP clock support --->
[*] GPIO Support --->
< > Dallas's 1-wire support ----

```

- SS_SATA_HOST:

```

-(-)
< > EMMC driver
<M> SStar SD/MMC Card Interface Support
< > EMAC
<M> Crypto driver
[ ] Enable AES DMA interrupt
<M> Support cryptodev
<*> Mstar CPU frequency scaling driver
-* MSTAR MIU
    MIU_PROTECT (ssc007a-s01a) --->
[*] Mstar BDMA driver
[ ] BDMA Line-Offset Enable
[*] Mstar Move DMA driver
< > SStar SATA HOST
< > SStar IR driver
<*> Sstar I2C driver
[ ] Support I2c ISR
<*> GPIO driver
< > SW I2C via GPIO support
[*] MS_RTC
< > Mstar RTC driver
<*> Mstar RTCWC driver
[ ] Mstar RTCPWC driver Error Handle on Hw Exception
<*> watchdog driver
< > sar driver
[*] MMA HEAP
-(+)

```

- Usb开关的config在kernel如下config

```
Device Drivers
ects submenus ---> (or empty submenus ----). Highlighted letters are hotkeys. Pressing <
Search. Legend: [*] built-in [ ] excluded <M> module < > module capable

+(-)
< > HSI support ----
  PPS support --->
  PTP clock support --->
[*] GPIO Support --->
< > Dallas's 1-wire support ----
[ ] Adaptive Voltage Scaling class support ----
[ ] Board level reset or power off ----
[ ] Power supply class support ----
< > Hardware Monitoring support ----
< > Generic Thermal sysfs driver ----
[*] Watchdog Timer Support --->
  Sonics Silicon Backplane --->
  Broadcom specific AMBA --->
  Multifunction device drivers --->
[ ] Voltage and Current Regulator Support ----
<*> Multimedia support --->
  Graphics support --->
< > Sound card support ----
  HID support --->
[ ] USB support ----
< > Ultra Wideband devices ----
<M> MMC/SD/SDIO card support --->
< > Sony MemoryStick card support ----
[ ] LED Support ----
[ ] Accessibility support ----
```

3. CPU单核运行400MHz

- CPU 以400MHz运行

kernel\arch\arm\boot\dts\infinity2m.dtsi

```

... cpus {
...     #address-cells = <1>;
...     #size-cells = <0>;

...     cpu@0 {
...         device_type = "cpu";
...         compatible = "arm,cortex-a7";
...         clock-frequency = <1000000000>;
...         clocks = <&CLK_cpupll_clk>;
...         reg = <0x0>;
...         operating-points = <
...             /* kHz      uV */
...             /*
...             1200000 1000000
...             1100000 1000000
...             1000000 900000
...             */
...             /*
...             800000 900000
...             600000 900000
...             */
...             400000 900000
...         >;
...     };
... };

```

- CPU单核运行

在bootargs最后加上nosmp即可：

```

bootargs=console=ttyS0,115200 root=/dev/mtdblock4
rootfstype=squashfs ro init=/linuxrc LX_MEM=0x3f00000

mma_heap=mma_heap_name0,miu=0,sz=0x1000000
mma_memblock_remove=1 highres=off

mmap_reserved=fb,miu=0,sz=0x300000,max_start_off=0x3300000,max_en
nosmp

```

公版的测试数据如下：

测试软件版本：TAKOYAKI_DLS00V009。

测试环境：Takoyaki NOR EVB板接TTL panel 12V供电。

		默认配置 (网络&usb 打开)	关闭网络	关闭网络 & usb	关闭网络 usb& sata
3.3V	IO	75.8	73	56.6	56.6
1.8V	AVDDIO_DRAM	32.1	32	32.02	32.02
	AVDDIO_DATA	49.5	50	49.4	49.5
0.9V	DVDD_DDR	7.54	7.5	7.52	7.52
	COREPOWER	127	124	120	120
	Total (unit: mA)	292	286	265.54	265.64



Check CPU temperature:

```
cat /sys/devices/virtual/mstar/msys/TEMP_R
```

Check CPU frequency:

```
cat sys/devices/system/cpu/cpu0/cpufreq/cpuinfo_cur_freq
```

